PEER REVIEW · BUILD-AGENT HANDOFF · v1.0

# agentic_fomprep — Autonomous Form Prep

Repository https://github.com/SimonBarnett/agentic_fomprep
Commit reviewed: `e9516f416fd5458ab25b8732f9a3139b277707ae` (main)
Claimed contract: CE_Priority_Autonomous_Form_Prep_Build_Plan.pdf v1.0
Stance: hostile. Scaffold-complete is not AT1-green.

**Do not mark MVP done. The park/restore SQL and DEV fail-closed gates are real. The web executor is an unverified selector sketch. AT4 cannot run on the actual CE estate (it demands global UPD=Y count == 3 while DEV has ~1000 unprepared forms). AT1 does not force UPD=Y. LASTPREPDATE is bigint; the "advanced" check can lie. A cookie file on disk is not a live Si session.**

## 0. Executive verdict

This is the best implementation of Hybrid F that exists in the org right now: module layout matches the plan, park is one transaction with a Y-count assert before commit, restore lives in finally, AllUnprepared is refused, WINRUN password is not in git, Playwright refuses Ignore Duplicate, PinComplete + frozen hosts block PRI. That is the good news.

The bad news is the pack has not proven a named form compiled on DEV. Off-DEV Test-Pack.ps1 only parses scripts, checks a JSON schema file, and runs AT6 + unit-mutex. Web selectors.json is tagged unverified-packaged-2026-09-16. formkeys.psd1 still contains KeyType='<PIN>'. Repo name is misspelled (fomprep). Disposition: ACCEPT as WP0–WP2 scaffold. REJECT as overnight runner until P0s below are green on CE-PRIORITY-DEV1.

| Layer | Status | Ship? |
|---|---|---|
| DEV freeze + PinComplete + mutex | Implemented | Yes |
| Park table + transactional park + finally restore | Implemented; AT4 brittle | Not until AT4 rewritten |
| CLI FORMPREP probe + tree kill | Implemented; password on argv | Demo only |
| Playwright Form Prep | Script exists; selectors unverified; frames unknown | NO |
| SQL verify UPD + LASTPREPDATE | Present; bigint compare is weak | Fix P0-V1 first |
| Error scrape e*msg / prep.err / report HTML | Wired; unproven on live UI | After first real run |
| FORMKEYS hook | Assert-only with placeholder dump | Assert skip until human dump |
| AT1 / AT2 / AT3 | AT1 does not match the plan; AT2/AT3 stubs | NO |
| Secrets | CredMan + log redact; argv still leaks | P1 |

## 1. Scope freeze for the next agent session

Allowed: P0 and P1 in this document, plus one live WhatIf and one AT4 rewrite on DEV1. Forbidden: AllUnprepared, FORMKEYS INSERT, Form Generator, REST discovery, renaming every file unless you also retarget remotes, running against PRI.

Definition of done: on CE-PRIORITY-DEV1, WhatIf prints three ZCLA exec ids; rewritten AT4 restores 5/5 without assuming global Y=3; one named form ends UPD=N with LASTPREPDATE changed from the pre-run snapshot; park rows all restored; screenshots in the run dir; no password in result.json or cli-stdout.txt.

## 2. What is already correct — do not regress

- Park INSERT+FLIP+COUNT inside one transaction; rollback on park_assert (New-ParkSnapshot.ps1).
- Refuse INSERT of a missing EXECPREPLOCK row (NOT NULL columns).
- Restore from SQL park table, not the txt dump. restoreOk requires OpenLeft=0 and FailedCount=0.
- finally always restores if $parked; restore short forces exit 3 even if targets compiled.
- Get-FrozenEnvironment.ps1 + AllowedComputer includes both CE-PRIORITY-DEV and CE-PRIORITY-DEV1.
- Scope AllUnprepared returns exit 2 without parking.
- WINRUN command logged as *** ; Protect-CliLog scrubs the secret if it landed in the file.
- Playwright dismisses Ignore/Yes-class dialogs (dangerousDialog regex) and unknown HTML modals as blocked-run.
- Web host check is prioritydev.clarksonevans.co.uk only.

- Install-OnDev refuses park DDL if OPEN park rows exist; drops wrong-type park table only when empty.

## 3. Findings — P0

**P0-T4** AT4 is impossible on the real CE estate **P0**

**Files.** tests/AT4-abort-restores.ps1 lines that throw if Get-YCount != 3, then expect 8 after forcing 5 dummies

**Problem.** AT4 requires global EXECPREPLOCK UPD='Y' count == 3 (the ZCLA trio only). The design problem this repo exists to solve is a 1000+ unprepared queue. On real DEV1 this test throws before it parks. If someone “fixes” the estate down to 3 so AT4 passes, they have hidden the production failure mode.

**Why it matters.** A build agent will either skip AT4, vandalise the queue to make Y=3, or declare WP1 done without crash-restore proof — the one test that matters.

**Fix.** Rewrite AT4: snapshot Y-count (no expected constant). Force 5 dummy N→Y. HoldPark on the three ZCLA names. Expect OPEN park rows == (Y_before + 5 - 3). Kill holder. RepairOpenParks. Assert those 5 dummies returned to their prev_upd. Assert ZCLA UPD/LASTPREPDATE unchanged. Restore any leftover OPEN in finally. Do not require Y_before==3.

**Acceptance.** AT4 PASS on DEV1 while Y-count is whatever the estate actually is (document the number in the log). Dummy five return to N. No nested park.

**P0-T1** AT1 does not implement the planned AT1 **P0**

**Files.** tests/AT1-already-prepared.ps1 ; build plan §16

**Problem.** Planned AT1: three already-prepared forms forced UPD=Y, runner → all UPD=N, LASTPREPDATE advanced, park restored. Actual AT1: call Prepare-Forms on the three names and hope. It does not force Y, does not snapshot LASTPREPDATE, does not assert FORMKEYS beyond a placeholder hook.

**Why it matters.** If the three ZCLA forms are already Y in the 1000-queue, you are testing “prep whatever is parked” not “re-prep these three.” If they are N and LASTPREPDATE is a same-day Priority bigint, Test-LastPrepAdvanced may fail or pass for the wrong reason.

**Fix.** Rewrite AT1: snapshot three rows; SET UPD='Y' (do not zero LASTPREPDATE unless -ResetLastPrepDate); run Prepare-Forms; assert each lastPrepDateAfter != before OR after > before using bigint compare; restoreOk; Y-count back to pre-test except the three now N.

**Acceptance.** AT1 PASS log prints before/after LASTPREPDATE integers for all three names.

**P0-V1** LASTPREPDATE compare is wrong for bigint Priority dates **P0**

**Files.** src/Private/Get-FormExec.ps1 Test-LastPrepAdvanced; config/dev.psd1 LastPrep=bigint

**Problem.** WP0 pinned LASTPREPDATE as bigint (0 = never). Test-LastPrepAdvanced tries datetime cast, then falls back to string inequality. Priority day-serial values often do not change when you re-prep the same form on the same calendar day. String '45221' -ne '45221' is false → form stays in stillUnprepared with a Blocker even though UPD flipped to N.

**Why it matters.** The hard rule “UPD=N AND LASTPREPDATE moved” will fail honest successful preps, or pass if any non-empty string differs (including garbage).

**Fix.** Compare as int64. Success if after > before, OR (before==0 AND after>0), OR (UPD became N AND after==before AND you passed -AllowSameDayPrep with an explicit flag and a warning). Default: require after > before. Log both integers.

**Acceptance.** Unit test: before=0 after=45221 true; before=45221 after=45221 false; before=45221 after=45222 true. No datetime parse.

**P0-W1** Web automation is not pinned to the real UI **P0**

**Files.** src/web/selectors.json pinnedAt=unverified-packaged-2026-09-16-pin-on-DEV1; src/web/formprep.mjs

**Problem.** Selectors are English getByText guesses. progressText is also 'Form Preparation' — the same string as the menu item — so progressSeen is set true immediately after the first OK click. The loop can mark completed 8 seconds later because “progress is gone.” Priority web is typically frames/iframes; getByText on page root often misses the prep dialog. First button named OK may be the wrong OK.

**Why it matters.** A green web-result.json with exitReason=completed and zero compiled forms is how this product lies.

**Fix.** On DEV1, record a headed run. Put real selectors (role+name, or frame locator) into selectors.json and set pinnedAt to the date they clicked Form Preparation for real. Wait for a progress control that is not the menu title. Do not set progressSeen=true on click alone.

**Acceptance.** One run directory with screenshots 01-after-goto … progress … errors-report.html AND SQL showing at least one target LASTPREPDATE change. If UI cannot be pinned in one working day, ship CLI-after-park only and leave web -SkipWeb default until pin.

**P0-A1** Cookie file is treated as a live session **P0**

**Files.** src/Private/Test-WebSession.ps1; Prepare-Forms.ps1 pre-park gate

**Problem.** Test-WebSession returns auth=ok if si-web-state.json exists and has cookies or origins. Stale cookies pass. The real login-page check is inside Playwright, which runs after park. Plan AT7: dead session → no park.

**Why it matters.** Overnight agent parks ~1000 rows, then hits login, then finally restore. Restore must work; if it does not, the estate is marked prepared. You invented the exact disaster the park table exists to prevent, every time the cookie file is stale.

**Fix.** Before park, run a 10s Playwright (or Invoke-WebRequest against an authenticated Priority URL) that fails closed on login title/url. storage_state_present is necessary, not sufficient. Keep the after-park login detector as a second line.

**Acceptance.** Delete/corrupt storageState → exit 2, zero rows inserted into AGENT_FORMPREP_PARK (AT7 as written in the plan).

**P0-I1** bigint vs int on exec id **P0**

**Files.** Get-FormExec.ps1 ExecId = [int]$r.exec_id ; New-ParkSnapshot uses [int64]; config ExecIdCol is bigint

**Problem.** T$EXEC is bigint. Casting to [int] overflows above 2,147,483,647. High custom exec ids on this estate already live in the 101883+ range (fits) but the cast is inconsistent and will corrupt park keys if any id grows.

**Why it matters.** Wrong exec id parked or restored is silent queue corruption.

**Fix.** Use int64 everywhere (result JSON schema already allows integer). Stop [int] casts on exec/pid.

**Acceptance.** Resolve ZCLA_* prints the same exec_id as SELECT T$EXEC FROM dbo.T$EXEC WHERE ENAME=...

## 4. Findings — P1

**P1-C1** WINRUN password is on the child process command line **P1**

**Files.** src/Private/Invoke-CliFormPrep.ps1 ProcessStartInfo.Arguments = user + password + paths

**Problem.** Any local admin, WMI Win32_Process, or Process Explorer sees the Si password for 60s. Protect-CliLog only scrubs the log file after exit.

**Why it matters.** DEV1 is a shared RDP box. This is how the secret leaves CredMan.

**Fix.** Prefer a response file / env var if winrun allows it. If not, document the argv leak as accepted DEV-only risk and keep CreateNoWindow. Never write Arguments to result.json (already avoided).

**Acceptance.** cli-stdout.txt contains *** only. A Process Explorer screenshot during the 60s window is the residual risk note in README.

**P1-L1** Test-LockIsLive pings COMPUTERNAME **P1**

**Files.** New-ParkSnapshot.ps1 Test-LockIsLive

**Problem.** A non-empty COMPUTERNAME with PID=0 is treated live if Test-Connection succeeds. Stale names that still DNS-resolve block all prep. ICMP is often firewalled so real live preps look stale.

**Why it matters.** False lock_held exit 2 overnight, or parking over a live Form Prep.

**Fix.** Live = (PID<>0 AND LOCKEXPIRY>0) only, matching the comment in the same file. Optional: if COMPUTERNAME equals this host and a winrun/formprep process exists. Drop ICMP.

**Acceptance.** Unit: PID=0 COMPUTERNAME=CE-PRIORITY-DEV → not live. PID=123 LOCKEXPIRY=1 → live.

**P1-R1** Restore does not put LOCKEXPIRY back **P1**

**Files.** sql/001 park table; Restore-ParkSnapshot.ps1

**Problem.** Park snapshot stores prev_upd/lastprep/computer/pid. Not LOCKEXPIRY. Restore leaves expiry at whatever the target-clear step or flip left.

**Why it matters.** Low. Still a lock-protocol drift vs Priority's own locker.

**Fix.** Add prev_lockexpiry to AGENT_FORMPREP_PARK (bigint) and restore it. Migrate only when OPEN count is 0 (Install-OnDev already refuses DDL on OPEN rows).

**Acceptance.** AT4 asserts dummy lockexpiry unchanged.

**P1-H1** FORMKEYS hook is a placeholder that looks configured **P1**

**Files.** hooks/formkeys.psd1 KeyType='<PIN>'; Apply-FormKeysRepair.ps1; AT1 -PostHooks FormKeysRepair

**Problem.** AT1 always passes FormKeysRepair. If Assert-FormKeysPresent treats <PIN> as skip, AT1 never proves keys survived. If it treats <PIN> as a required row, AT1 always exit 4.

**Why it matters.** Build agent will "fix" hooks by inventing INSERT statements — forbidden in MVP.

**Fix.** Empty arrays for all three ZCLA forms until a human pastes a dump. Assert-FormKeysPresent must no-op (return true) when every Col/KeyType is missing or '<PIN>'. README: hook waived until dump date.

**Acceptance.** AT1 without a dump does not fail on hooks. After a real dump, missing PART key → exit 4.

**P1-W2** node missing silently skips web after park **P1**

**Files.** Invoke-WebFormPrep.ps1 returns Status=skipped if node is absent

**Problem.** Prepare-Forms will park, CLI-timeout, web-skip, then verify. Targets stay Y → exit 3 after restore. Overnight you get a restore drill and no prep, with a skip buried in web-result absence.

**Why it matters.** Looks like Form Prep failed when the box just has no Node.

**Fix.** If -not SkipWeb and node missing: do not park (detect before park) or fail exit 2 node_missing. Install-OnDev already warns; make it a hard check in Assert-Environment when SkipWeb is false.

**Acceptance.** AT on a PATH without node: WhatIf OK; real run without SkipWeb exits 2, parkedCount=0.

**P1-S1** Test-WebSession + SkipWeb bypasses auth gate **P1**

**Files.** Prepare-Forms.ps1 auth_expired only when -not SkipWeb

**Problem.** CLI-only path parks with no session file. Fine if CLI is the executor; still needs the park/restore discipline. Document it. AT7 must run without -SkipWeb.

**Why it matters.** AT7 as currently skipped off-DEV will be run with SkipWeb by a lazy agent.

**Fix.** AT7 script: no SkipWeb, missing storageState, assert parkedCount=0.

**Acceptance.** tests/AT7-auth-expired-no-park.ps1 fails if any INSERT hit AGENT_FORMPREP_PARK.

## 5. Findings — P2

**P2-N1** Repository name is a typo **P2**

**Files.** SimonBarnett/agentic_fomprep

**Problem.** fomprep not formprep. Every clone path, Grok worktree, and human search will miss it.

**Why it matters.** Noise and onboarding cost.

**Fix.** Leave GitHub name if remotes are painful; set description “Priority Form Prep (spelling: formprep)”. Do not spend the MVP night on a rename.

**Acceptance.** README first line contains both spellings.

**P2-T2** AT2/AT3 are not tests **P2**

**Files.** tests/AT2-broken-trigger.ps1 ; tests/AT3-index-dialog.ps1

**Problem.** Stubs. README already allows waive. Keep them waived with a one-line reason and a ticket, or they will be “skipped” forever while claiming coverage.

**Why it matters.** False sense of dialog/trigger safety.

**Fix.** AT3: feed formprep.mjs a fixture HTML modal containing Ignore Duplicate Values; assert action=blocked-run and no click. Does not need DEV.

**Acceptance.** tools/Test-Pack.ps1 runs that fixture test off-DEV.

**P2-G1** WhatIf ok=false exit 0 surprises agents **P2**

**Files.** Prepare-Forms.ps1 WhatIf branch; README table

**Problem.** Documented, but Complete-ResultObject + callers that treat ok as the only signal will page.

**Why it matters.** Low.

**Fix.** Keep exit 0. Set reason=whatIf and print it. Test-Pack already can assert this.

**Acceptance.** WhatIf JSON whatIf=true ok=false exitCode=0 parkedCount=0.

## 6. Ordered work package

- **Step 0.** On DEV1 only: Prepare-Forms -WhatIf for the three ZCLA names. Paste exec ids and current Y-count into the PR.
- **Step 1.** P0-V1 bigint LASTPREPDATE compare + unit cases in tests/.
- **Step 2.** P0-T4 rewrite AT4 (no global Y==3). Run it. Paste Y-count before/after.
- **Step 3.** P0-A1 live session probe before park. AT7 green.
- **Step 4.** P0-T1 rewrite AT1. Run once. If web selectors fail, rerun AT1 -SkipWeb after a parked CLI that actually moves LASTPREPDATE; if CLI cannot move it, stop and pin selectors (P0-W1) — do not flip UPD by SQL.
- **Step 5.** P0-I1 int64 everywhere.
- **Step 6.** P1-L1 drop ICMP. P1-W2 node_missing before park. P1-H1 empty hook dump.
- **Step 7.** Off-DEV: fixture AT3 modal in Test-Pack.ps1.
- **Step 8.** Only then default to web+CLI together on a weekday with a human watching the first headed Playwright.

## 7. Regression tests the agent must add

| Test | Asserts |
|---|---|
| unit.lastprep.bigint | 0→N true; same-day equal false; N+1 true; no DateTime |
| unit.lock.no-icmp | PID=0 + computer set → not live |
| unit.execid.int64 | id 3_000_000_001 does not overflow |
| AT4.estate-independent | PASS when Y-count is not 3 |
| AT7.no-park | missing storageState, SkipWeb false, park table insert count 0 |
| AT1.force-Y | explicit UPDATE Y then prep; prints before/after integers |
| pack.modal-fixture | Ignore Duplicate Values → blocked-run |
| whatIf.no-write | WhatIf leaves AGENT_FORMPREP_PARK rowcount unchanged |

## 8. Residual operational risk

Even after P0s: Form Prep on CE still surfaces index rebuild dialogs that this runner must not click. Park remains Option A applied to everyone else — a failed restore is still Sev-1 on DEV. WINRUN on argv leaks Si on a shared box. Playwright headed needs an interactive session or an unlocked RDP desktop overnight. Git metadata for this repo on M: is broken; keep separate-git-dir on C: as already done for the working copy.

Do not “help” AT4 by preparing the entire unprepared estate down to three rows. That is how you ship a test and unship the product.

## 9. Review metadata

| Field | Value |
|---|---|
| Repo | https://github.com/SimonBarnett/agentic_fomprep |
| Commit | e9516f416fd5458ab25b8732f9a3139b277707ae |
| Plan | CE_Priority_Autonomous_Form_Prep_Build_Plan.pdf v1.0 (in repo root) |
| Pin | config/dev.psd1 PinComplete=true; SqlDatabase=system; ExecTable=dbo.T$EXEC |
| Not executed here | No DEV1 SQL, no Playwright, no AT4 live |
| Disposition | REJECT as overnight MVP. ACCEPT as Hybrid F scaffold if P0-T4/T1/V1/W1/A1/I1 are scheduled. |

Next artefact from the build agent: a PR whose commits cite finding IDs (fix P0-T4 AT4 without Y==3). Not a new executor. Not AllUnprepared.